

CLASS implementation note: scalar-field couplings to matter and dark radiation

Branch DE_DR_DM_int

1 Purpose and conventions

This note summarizes the CLASS implementation of scalar-field couplings to matter and dark radiation, following the conventions of `REFS_FOR_CLAUDE/NotesRadiationCoupling260616.tex`. The paper example for the matter-coupled model is `2505.10410v2.pdf`; the two radiation-coupled models are those of `NotesRadiationCoupling_DONTTOUCH.pdf`.

We use a scalar field ϕ with potential $V(\phi)$ and multiplicative coupling functions $A_m(\phi)$ and $A_r(\phi)$ such that

$$\rho_m = A_m(\phi) \bar{\rho}_m, \quad \rho_r = A_r(\phi) \bar{\rho}_r, \quad \bar{\rho}_m \propto a^{-3}, \quad \bar{\rho}_r \propto a^{-4}. \quad (1.1)$$

The fiducial background values used in the comparison runs are

$$\Omega_{r0} = 10^{-4}, \quad \Omega_{m0} = 0.3149, \quad \Omega_{\phi0} = 0.6850, \quad \Omega_{k0} = 0. \quad (1.2)$$

All implementation changes are inside CLASS. The external scripts used during development only read CLASS output tables and produced the figures included below.

2 Matter-coupled scalar background

For the matter sector, the new form is

$$A_m(\phi) = 1 + C_m \left(1 - e^{-\beta_m \phi}\right), \quad C(\phi) = A_m(\phi)^2. \quad (2.1)$$

In the code this is selected by

$$\text{scf_conformal_form} = \text{matter_exp}. \quad (2.2)$$

The existing qcdm–scalar conformal machinery is then reused through `C_scf`, `dC_scf`, `ddC_scf`, `Q_scf`, and the existing perturbed `delta_Q_scf` coefficients.

The option

$$\text{scf_ic_from_today} = \text{yes} \quad (2.3)$$

reconstructs ϕ_{ini} , ϕ'_{ini} , V_0 , and the initial qcdm density from the requested today values. This is used by `fig3_2505_matter.ini`. The slope used for the Fig. 3 reproduction is

$$\lambda = \sqrt{2}, \quad (2.4)$$

as in Eq. (3.8) of `2505.10410v2.pdf`.

For this case the effective dark-energy variables are

$$\rho_{\text{DE}} = \rho_\phi + (A_m - 1)\bar{\rho}_m, \quad w_{\text{DE}} \rho_{\text{DE}} = p_\phi. \quad (2.5)$$

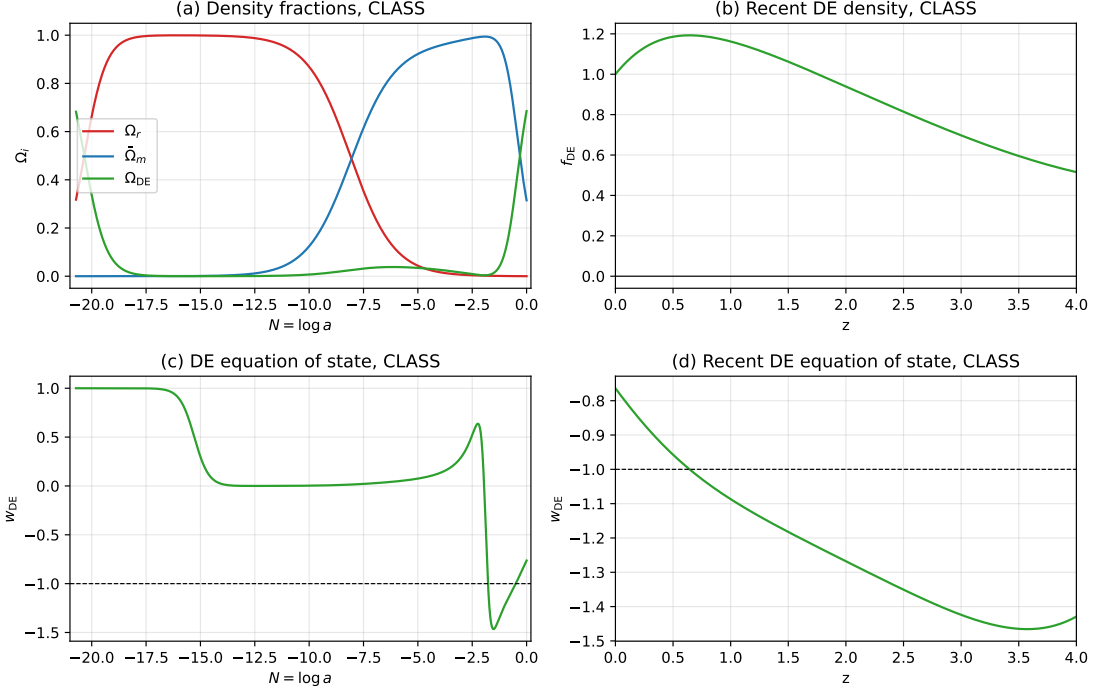


Figure 1: Matter-coupled CLASS reproduction for the `fig3.2505_matter.ini` setup. The plotted matter fraction uses $\bar{\rho}_m = \rho_{\text{qcdm}}/A_m$.

At $z = 0$, the CLASS output gives

quantity	CLASS value
$w_{\phi 0}$	-0.763660007
w_{DE0}	-0.763660007
Ω_{r0}	9.9999997×10^{-5}
Ω_{m0}	0.314900009
Ω_{DE0}	0.684999991
ϕ_0	-1.31×10^{-8}

3 Radiation-coupled scalar background

The radiation coupling is independent of the qcdm conformal factor. It is selected by

$$\text{scf_interacting_radiation} = \text{yes} \quad (3.1)$$

and uses the free parameters `scf_C0.r` and `scf_beta.r`:

$$A_r(\phi) = 1 + C_r \left(1 - e^{-\beta_r \phi} \right). \quad (3.2)$$

The CLASS background density stored in `rho_idr` is the full coupled density,

$$\rho_{\text{idr}} = A_r(\phi) \bar{\rho}_{\text{idr}}. \quad (3.3)$$

The pressure is kept as radiation,

$$p_{\text{idr}} = \frac{1}{3}\rho_{\text{idr}}. \quad (3.4)$$

The normalization of the radiation Lagrangian only fixes the field amplitude inside $\bar{\rho}_{\text{idr}}$. The implemented object is the radiation stress tensor,

$$T^0_0 = -\rho_{\text{idr}}, \quad T^i_j = \frac{1}{3}\rho_{\text{idr}}\delta^i_j, \quad (3.5)$$

so A_r rescales the full dark-radiation tensor. The anisotropic-stress hierarchy is otherwise the existing CLASS idr hierarchy.

The scalar background equation is implemented in CLASS density units as

$$\phi'' + 2\mathcal{H}\phi' + a^2(V_{,\phi} - Q_m - Q_r) = 0, \quad Q_r = -3\rho_{\text{idr}}\frac{\partial_\phi A_r}{A_r}. \quad (3.6)$$

This sign matches the physical equation

$$\ddot{\phi} + 3H\dot{\phi} + V_{,\phi} + \bar{\rho}_m\partial_\phi A_m + \bar{\rho}_r\partial_\phi A_r = 0, \quad (3.7)$$

after using the CLASS convention in which densities are multiplied by $8\pi G/3$.

The effective dark-energy output is

$$\rho_{\text{DE}} = \rho_\phi + (A_m - 1)\bar{\rho}_m + (A_r - 1)\bar{\rho}_r, \quad (3.8)$$

$$w_{\text{DE}}\rho_{\text{DE}} = p_\phi + \frac{1}{3}(A_r - 1)\bar{\rho}_r. \quad (3.9)$$

Radiation coupling: reproduced with Fig. 1 axes from NotesRadiationCoupling

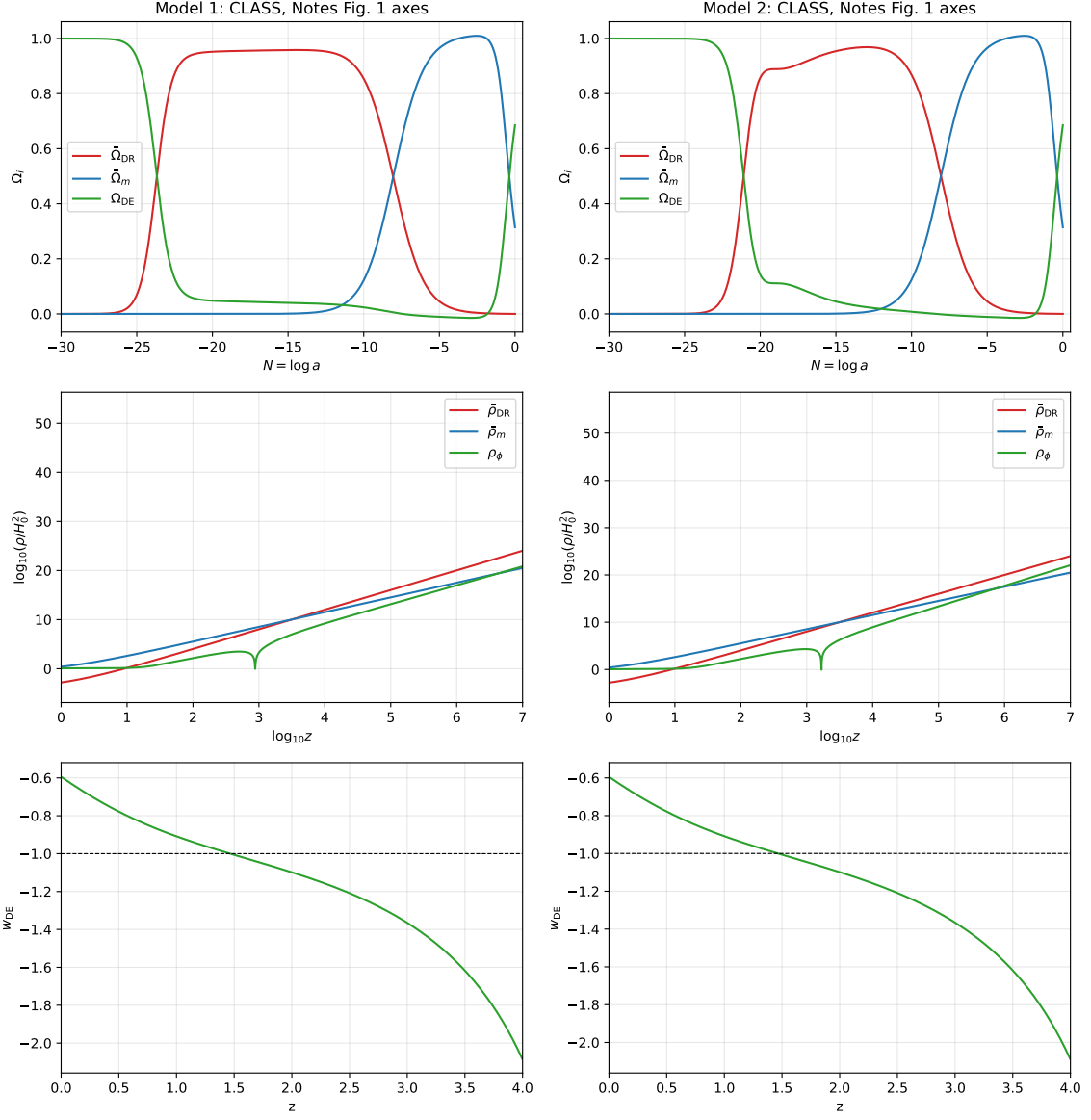


Figure 2: Radiation-coupled CLASS reproduction for the two NotesRadiationCoupling Fig. 1 models. The plotted barred radiation and matter densities are reconstructed as $\bar{\rho}_r = \rho_{\text{idr}}/A_r$ and $\bar{\rho}_m = \rho_{\text{qcdm}}/A_m$.

At $z = 0$, the two Notes models give

quantity	model 1	model 2
$w_{\phi 0}$	-0.594208876	-0.594223632
$w_{\text{DE}0}$	-0.594208863	-0.594223633
Ω_{r0}	1.0000022×10^{-4}	9.9999990×10^{-5}
Ω_{m0}	0.314899728	0.314900018
$\Omega_{\text{DE}0}$	0.685000288	0.684999981
ϕ_0	1.30×10^{-6}	-5.69×10^{-8}
$A_r(\phi_0)$	0.999999901	1.000000001

4 Radiation perturbations

The perturbations are implemented for the full coupled idr variables

$$\delta_{\text{idr}} \equiv \frac{\delta \rho_{\text{idr}}}{\rho_{\text{idr}}}, \quad \theta_{\text{idr}}. \quad (4.1)$$

Define

$$\Lambda_r \equiv \frac{\partial_\phi A_r}{A_r}, \quad \Lambda_{2,r} \equiv \frac{d^2 \ln A_r}{d\phi^2} = \frac{\partial_\phi^2 A_r}{A_r} - \Lambda_r^2. \quad (4.2)$$

In synchronous gauge, the added idr terms are

$$\delta'_{\text{idr}} = -\frac{4}{3} \left(\theta_{\text{idr}} + \frac{h'}{2} \right) + \Lambda_r \delta \phi' + \Lambda_{2,r} \phi' \delta \phi, \quad (4.3)$$

$$\theta'_{\text{idr}} = \frac{k^2}{4} \delta_{\text{idr}} - k^2 \sigma_{\text{idr}} + \frac{3}{4} \Lambda_r k^2 \delta \phi - \Lambda_r \phi' \theta_{\text{idr}}. \quad (4.4)$$

IDM–DR scattering terms, when present, are preserved and the scalar force is added on top of the usual idr Euler equation.

The scalar perturbation equation receives

$$\delta Q_r = -3\rho_{\text{idr}} (\Lambda_r \delta_{\text{idr}} + \Lambda_{2,r} \delta \phi), \quad (4.5)$$

which enters the perturbed Klein–Gordon equation as $a^2 \delta Q_r$, analogously to the existing qcdm `delta_Q_scf` term. In Newtonian gauge the metric term $2\psi Q_r$ is added in the same way as for the qcdm source.

5 Code changes: before and after

This section documents the concrete code edits behind the equations above, as *before/after* pairs taken from the three implementation commits (11432659, 2f4abc0d, dfb1975c) relative to the pre-implementation state (74a2ed72). Line numbers refer to the current tree. Blocks marked “new” had no prior counterpart.

5.1 source/background.c: matter conformal form

The conformal factor selector gained the `matter_exp` branch $C(\phi) = A_m(\phi)^2$ with $A_m = 1 + C_m(1 - e^{-\beta_m \phi})$ [Eq. (2.1)]. Shown for `C_scf`; `dC_scf` and `ddC_scf` were extended identically.

Before (`C_scf`, only the exponential form existed):

```

switch (pba->scf_conformal_form) {
case scf_conformal_exp:
    return scf_C0*exp(2.*scf_beta*phi);
default:
    class_stop(pba->error_message, "Unknown scf_conformal_form %d", pba->scf_conformal_form);
    return 1.0;
}

```

After (new matter_exp case, plus a shared helper):

```

static void scf_matter_exp_A(struct background *pba, double phi,
                           double *A, double *dA, double *ddA) {
    double Aphi = pba->C0_scf, alpha = pba->beta_scf, e = exp(-alpha*phi);
    *A = 1.0 + Aphi*(1.0 - e);
    *dA = Aphi*alpha*e;
    *ddA = -Aphi*alpha*alpha*e;
}
/* ... inside C_scf: */
switch (pba->scf_conformal_form) {
case scf_conformal_exp:
    return scf_C0*exp(2.*scf_beta*phi);
case scf_conformal_matter_exp: /* NEW */
    scf_matter_exp_A(pba, phi, &A, &dA, &ddA);
    return A*A;
default:
    class_stop(pba->error_message, "Unknown scf_conformal_form %d", pba->scf_conformal_form);
    return 1.0;
}

```

5.2 source/background.c: radiation coupling function (new)

The multiplicative radiation factor A_r and its derivatives [Eq. (3.2)] were added with the same exponential template:

New (source/background.c, ~1.3661--3715):

```

static void scf_radiation_exp_A(struct background *pba, double phi,
                               double *A, double *dA, double *ddA) {
    double Aphi = pba->C0_r_scf, beta = pba->beta_r_scf, e = exp(-beta*phi);
    *A = 1.0 + Aphi*(1.0 - e);
    *dA = Aphi*beta*e;
    *ddA = -Aphi*beta*beta*e;
}
double A_r_scf(...) { ... return A; } /* A_r(phi) */
double dA_r_scf(...) { ... return dA; } /* d A_r / dphi */
double ddA_r_scf(...) { ... return ddA; } /* d^2 A_r / dphi^2 */

```

5.3 source/background.c: coupled rho_idr

The interacting-dark-radiation density is multiplied by $A_r(\phi)$ so that `rho_idr` stores the full coupled density [Eq. (3.3)].

Before (~1.630):

```

if (pba->has_idr == _TRUE_) {
    pvecback[pba->index_bg_rho_idr] = pba->Omega0_idr * pow(pba->H0,2) / pow(a,4);
    rho_tot += pvecback[pba->index_bg_rho_idr];
    ...
}

```

After:

```

if (pba->has_idr == _TRUE_) {
  pvecback[pba->index_bg_rho_idr] = pba->Omega0_idr * pow(pba->H0,2) / pow(a,4);
  if (pba->has_scf_radiation == _TRUE_) { /* NEW */
    pvecback[pba->index_bg_rho_idr] *= A_r_scf(pba,phi); /* NEW */
  }
  rho_tot += pvecback[pba->index_bg_rho_idr];
  ...
}

```

5.4 source/background.c: scalar background equation

The background Klein–Gordon source $-Q$ in $\phi'' + 2\mathcal{H}\phi' + a^2(V_\phi - Q) = 0$ gained the radiation term $Q_r = -3\rho_{\text{idr}} \partial_\phi A_r / A_r$ [Eq. (3.6)].

Before (~ 1.3105):

```

if (pba->has_scf == _TRUE_) {
  double Q_bg_scf = 0.;
  if (pba->has_qcdm_de_q == _TRUE_) {
    Q_bg_scf = Q_scf(pba, y[..phi], y[..phi_prime], y[..rho_qcdm], a, pvecback);
  }
  dy[pba->index_bi_phi_scf] = y[pba->index_bi_phi_prime_scf]/a/H;
  ...
}

```

After:

```

if (pba->has_scf == _TRUE_) {
  double Q_bg_scf = 0.;
  if (pba->has_qcdm_de_q == _TRUE_) {
    Q_bg_scf = Q_scf(pba, y[..phi], y[..phi_prime], y[..rho_qcdm], a, pvecback);
  }
  if (pba->has_scf_radiation == _TRUE_) { /* NEW */
    double A_r = A_r_scf(pba, y[pba->index_bi_phi_scf]);
    class_test(A_r <= 0., error_message, "A_r(phi) = %e must stay positive.", A_r);
    /* rho_idr stores A_r rho_bar_r; CLASS units give Q_r = -3 rho_bar_r A_r,phi */
    Q_bg_scf += -3.0*(pvecback[pba->index_bg_rho_idr]/A_r)*dA_r_scf(pba, y[pba->index_bi_phi_scf]);
  }
  dy[pba->index_bi_phi_scf] = y[pba->index_bi_phi_prime_scf]/a/H;
  ...
}

```

5.5 source/background.c: effective dark-energy output (new)

The output block was extended to print ρ_{DE} and w_{DE} including the radiation term [Eqs. (3.8)–(3.9)]:

New (~ 1.2924):

```

double A_r = (pba->has_scf_radiation==_TRUE_) ? pvecback[pba->index_bg_A_r_scf] : 1.0;
double rho_bar_r = (pba->has_scf_radiation==_TRUE_) ? pvecback[pba->index_bg_rho_idr]/A_r : 0.0;
double matter_coupling = 0.0;
if (pba->has_qcdm_de_q == _TRUE_) {
  double A_scf = sqrt(pvecback[pba->index_bg_C_scf]); /* A_m = sqrt(C) */
  matter_coupling = pvecback[pba->index_bg_rho_qcdm]*(1.0 - 1.0/A_scf);
}
double rho_de_eff = pvecback[pba->index_bg_rho_scf] + matter_coupling + (A_r-1.0)*rho_bar_r;
double p_de_eff = pvecback[pba->index_bg_p_scf] + (1.0/3.0)*(A_r-1.0)*rho_bar_r;

```

5.6 source/perturbations.c: coupling coefficients (new)

Λ_r , $\Lambda_{2,r}$ and Q_r [Eq. (4.2)] are precomputed once per call:

New (~ 1.9365):

```
if (use_radiation == _TRUE_) {
    lambda_r_scf = pvecback[pba->index_bg_dA_r_scf]/pvecback[pba->index_bg_A_r_scf];
    lambda2_r_scf = pvecback[pba->index_bg_ddA_r_scf]/pvecback[pba->index_bg_A_r_scf]
        - lambda_r_scf*lambda_r_scf;
    Q_r_scf = -3.0*pvecback[pba->index_bg_rho_idr]*lambda_r_scf;
}
```

5.7 source/perturbations.c: idr continuity and Euler

The two source terms of Eqs. (4.3)–(4.4) are added on top of the standard idr fluid equations.

Before (~ 1.9979):

```
dy[pv->index_pt_delta_idr] = -4./3.*(theta_idr + metric_continuity);
...
dy[pv->index_pt_theta_idr] = k2*(y[pv->index_pt_delta_idr]/4.-s2_squared*y[pv->index_pt_shear_idr]) +
    metric_euler;
```

After:

```
dy[pv->index_pt_delta_idr] = -4./3.*(theta_idr + metric_continuity);
if (use_radiation == _TRUE_) { /* NEW */
    dy[pv->index_pt_delta_idr] +=
        lambda_r_scf*y[pv->index_pt_phi_prime_scf]
        + lambda2_r_scf*pvecback[pba->index_bg_phi_prime_scf]*y[pv->index_pt_phi_scf];
}
...
dy[pv->index_pt_theta_idr] = k2*(y[pv->index_pt_delta_idr]/4.-s2_squared*y[pv->index_pt_shear_idr]) +
    metric_euler;
if (use_radiation == _TRUE_) { /* NEW */
    dy[pv->index_pt_theta_idr] +=
        0.75*lambda_r_scf*k2*y[pv->index_pt_phi_scf]
        - lambda_r_scf*pvecback[pba->index_bg_phi_prime_scf]*theta_idr;
}
```

5.8 source/perturbations.c: perturbed Klein–Gordon source

The radiation source δQ_r [Eq. (4.5)] is built and injected into the perturbed KG equation as $a^2\delta Q_r$, with the Newtonian-gauge $2\psi Q_r$ term, mirroring the qcdm `delta_Q_scf` source.

Before (synchronous KG, no radiation term):

```
dy[pv->index_pt_phi_prime_scf] =
    -2.*a_prime_over_a*y[pv->index_pt_phi_prime_scf]
    -(k2+a2*pvecback[pba->index_bg_ddV_scf])*y[pv->index_pt_phi_scf]
    -metric_continuity*pvecback[pba->index_bg_phi_prime_scf]
    + kg_term_entropy + kg_term_q;
```

After (source assembled, then added):

```
if (use_radiation == _TRUE_) { /* NEW */
    delta_Q_r_scf = Q_r_scf*delta_idr
        - 3.0*pvecback[pba->index_bg_rho_idr]*lambda2_r_scf*y[pv->index_pt_phi_scf];
}
...
double kg_term_r = 0.;
```



```

if (use_radiation == _TRUE_) { /* NEW */
  kg_term_r = a2*delta_Q_r_scf;
  if (ppt->gauge == newtonian)
    kg_term_r += 2.*a2*pvecmetric[ppw->index_mt_psi]*Q_r_scf;
}
dy[pv->index_pt_phi_prime_scf] =
  -2.*a_prime_over_a*y[pv->index_pt_phi_prime_scf]
  -(k2+a2*pvecback[pba->index_bg_ddV_scf])*y[pv->index_pt_phi_scf]
  -metric_continuity*pvecback[pba->index_bg_phi_prime_scf]
  + kg_term_entropy + kg_term_q + kg_term_r; /* + kg_term_r NEW */

```

5.9 source/input.c and include/background.h

Parsing of the new switches and storage of the new parameters/indices:

- include/background.h: added flags `has_scf_radiation`, parameters `C0_r_scf`, `beta_r_scf`, the enum value `scf_conformal_matter_exp`, and the background indices `index_bg_A_r_scf`, `index_bg_dA_r_scf`, `index_bg_ddA_r_scf`.
- source/input.c: parsing of `scf_interacting_radiation`, `scf_C0_r`, `scf_beta_r`, and the `scf_conformal_form = matter_exp` keyword, together with the `scf_ic_from_today` reconstruction used by the Fig. 3 setup.

6 Implemented files and validation

The implementation is distributed over the following files:

- include/background.h: new radiation-coupling parameters and background indices.
- source/input.c: parsing of `scf_interacting_radiation`, `scf_C0_r`, and `scf_beta_r`.
- source/background.c: A_r , $\partial_\phi A_r$, $\partial_\phi^2 A_r$, coupled `rho_idr`, scalar background source, and effective dark-energy outputs.
- source/perturbations.c: idr continuity/Euler source terms and the radiation contribution to the perturbed scalar equation.
- fig3_2505_matter.ini, notes_fig1_model1.ini, and notes_fig1_model2.ini: reproduction configurations.

The following checks were run:

- make class
- ./class fig3_2505_matter.ini
- ./class notes_fig1_model1.ini
- ./class notes_fig1_model2.ini

The perturbation code was additionally smoke-tested with a temporary default-like cosmology containing a small scalar-field density and a small idr density, using `output = mPk`. That run completed the background, thermodynamics, source, primordial, and Fourier modules and wrote a matter power spectrum.

The two comparison figures are regenerated directly from the CLASS background tables by the self-contained script `doc/figures/make_comparison_figs.py`, which reads `output/fig3_2505_matter_b`, `output/notes_fig1_model1_background.dat`, and `output/notes_fig1_model2_background.dat` and writes the `.pdf` and `.svg` versions of both figures. Rerunning `make class`, the three `.ini` files above and that script reproduces the $z = 0$ tables and the figures shown here.

7 Comments and limitations

The exact Notes Fig. 1 configurations are designed for background comparisons: they use `Omega_g = 1.e-12` and a very large `xi_idr` so photons are only a trace component while `idr` carries the desired radiation budget. This is not a healthy thermodynamics setup for CMB perturbation tests, so perturbative stability was tested on a default-like temporary configuration instead.

The current radiation-coupling function is the exponential form (3.2). The parameters are free, but adding other functional forms would require extending `A_r_scf`, `dA_r_scf`, and `ddA_r_scf` in `source/background.c`.